

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (3rd Semester, CST) Final Semester Examination, November 2024

Digital Logic (CS 2102)

F.M. 50

Time: 3 hours

Answer any 5 questions

1. Define lockout of counters. Design a mod-5 synchronous counter using JK flip-flop and minimum number of external gates so that if the unused states 101, 110 and 111 occur, the next clock pulse will reset the counter to 000. [2 + 8]

2. A BCD-to-seven-segment decoder is a combinational circuit that accepts a decimal digit in BCD and generates the appropriate outputs for selection of segments in the display indicator used for displaying the decimal digit. The seven outputs of the decoder (a, b, c, d, e, f, g) select the corresponding segments in the display as shown below. Design the BCD-to-seven-segment decoder circuit. [10]

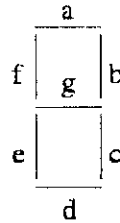


Figure 1: Segment designation.

3. Show that an 8-to-1 multiplexer can be implemented using 2-to-1 multiplexer modules. Hence, implement the following function using above mentioned *MUX* and at most one addition gates.

$$F(A, B, C, D) = \overline{A} \overline{B} C + A \overline{B} \overline{D} + BC$$

[5 + 5]

4. (a) Define Moor machine and Mealy machine.

Consider the following state table of a Moore Machine and find the equivalent states. Here, *PS* $\Rightarrow$ Present state, *NS* $\Rightarrow$ Next state, and *x* $\Rightarrow$ Input. You have to use one implication chart for each iteration. [2 + 8]

PS	NS		Output
	x = 0	x = 1	
A	D	C	0
B	F	H	0
C	E	D	1
D	A	F	0
E	C	A	1
F	F	B	1
G	B	H	0
H	C	G	1

5. (a) Draw the circuit diagram of a 4-bit bidirectional shift register and explain its operation.

(b) A circuit accepts a serial bit stream X as input and produces a serial bit Z as output. Whenever the bit pattern "0110" appears in the input stream, its output $Z = 1$ and at all other times, $Z = 0$. Overlapping occurrences of the patterns are also detected. Draw the state diagram of the circuit. [5 + 5]

6. Consider the following Boolean function in four variables, where m indicates the minterms for which f is true and d denotes the don't care minterms.

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 10, 12) + \sum d(4, 7, 8, 14)$$

You are given only one NOT gate and unlimited supply of 2-input AND and OR gates. Only the variables A , B , C and D are available as inputs but their compliments are not. Your task is to realize F under the given constraints.

(a) Draw the K-map of the complement function $\overline{F}$.

(b) From the K-map, obtain $\overline{F}$ in such a manner that the expression does not have any complimented variable.

(c) Draw the logic circuit diagram for F (not $\overline{F}$). [5 + 2 + 3]

7. (a) Define the following terms: *Fan-out*, and *Noise Margin*.

(b) Draw the electronic circuit diagram of a 2-input DTL NAND gate and explain its operation. Explain which logic level limits the fan-out of DTL logic gates. [3 + (5 + 2)]

8. Short Notes on:

(a) Johnson Counter

(b) Open collector TTL [4 + 6]